



IDENTIFICATION

CODE : GI-5-S1-EC-OTL
ECTS : 3

HOURS

Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

- A2 Operate a model of a real or virtual system (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C5 Managing a production system and react to malfunctions (Level 2)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)
- C13 Considering technological and methodological innovation (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

To be able to :

CONTENT

The engineering student will work and be assessed on the following knowledge and skills :

- (10h) Know the "classical" transportation problems : Traveling Salesman Problem (TSP), Vehicle Routing Problem (VRP), etc.
- (2h) Know the basic methods and tools to design logistic networks.
- (8h) Know the principles and tools for distributed decision-making: game theory, multi-agent systems and agent-based simulation coupled with flow simulation (for instance with Anylogic)
- (4h) Know the tools (especially decomposition methods) to perform joint optimisation.
- Be able to formulate and solve a supply chain optimisation problem, from the viewpoint of transportation and flow management.

BIBLIOGRAPHY

- Ilya Grigoryev. AnyLogic 7 in Three Days : A Quick Course in Simulation Modeling. 3rd edition.
- Jacques Teghem. Recherche opérationnelle. Tome 1, Tome 1, OCLC : 819196866. Paris : Ellipses, 2012. isbn : 978-2-7298-7509-1.
- Paolo Toth et Daniele Vigo, eds. Vehicle routing : problems, methods, and applications. Second edition. MOS-SIAM series on optimization. Philadelphia : Society for Industrial et Applied Mathematics : Mathematical Optimization Society, 2014. 463 p. isbn : 978-1-61197-358-7.
- Michael Wooldridge. An Introduction to Multi-Agent Systems. 2d edition. New York, NY, USA: John Wiley & Sons, Inc., 2009.
- Murat Yildizoglu. Introduction à la Théorie des Jeux. Dunod, Paris (France), 2003.

PRE-REQUISITES

Inventory management : EOQ, basic replenishment methods
Flow management
Production management : S&OP, MPS & MRP
Scheduling: models, standard scheduling notation, heuristic solving methods for scheduling problems
Combinatorial optimisation: graph theory
Logistic

